

Assessment of the National Early Warning Alert and Response (EWAR) System with a Focus on Event-Based Surveillance (EBS)

Developed by the US CDC Global Surveillance, Laboratory, and Data Sciences Branch

Date Survey Completed:

Name of the Country:

Ministry focal points/partners interviewed:

Name	Title/Position	Organization	Email Address	Date Interviewed

Additional sectors/partners supporting EWAR and EBS implementation in country:

Name	Title/Position	Organization	Email Address
		E.g., WHO, IFRC, Taskforce for Global Health	

Survey Facilitator/Translator:

Name	Title/Position and survey role	Organization	Email Address	Language

What is the purpose of this tool?

The objective of this assessment is to achieve a basic understanding of existing surveillance structures that are currently supporting or could support early warning alert and response (EWAR) activities, with a focus on event-based surveillance (EBS). ***This specific assessment is meant for countries that already have EBS in place.***

Who should complete and/or participate in this assessment?

Ministry of Health (MoH) surveillance staff, including but not limited to EBS focal points, surveillance officers, public health analysts, and Emergency Operations Center staff, should participate in this assessment. For countries implementing a multisectoral, One Health or all-hazards approach to EBS, sectors covering animal, environment, and border health may also be relevant to include.

How will information be collected, shared, and use?

This assessment can be self-administered or facilitated by U.S. CDC and their implementing partners. Assessment outcomes will not be shared outside of CDC, unless approved by the MoH/National Public Health Institute (NPHI) and any other participating sectors. Outcomes will help identify potential areas for technical support and aid in the design of tailored, integrated, surveillance platforms in country that include EWAR and EBS. Repeating assessments may help track progress and changes in capacity seen over time. And this assessment can help provide the system background needed to inform a more formal surveillance evaluation.

Questions

Surveillance & Detection

- 1. Do you have a list of priority diseases, syndromes, and conditions under the country's current national surveillance?** Yes/No/Don't know
 - a. If yes, how many diseases/syndromes/conditions are on this list?
 - b. If yes, when was the last time this list was revised?
 - c. If yes, do you have a copy of this list or something to help guide you on the priority diseases/syndromes/conditions? Yes/No
 - i. If yes, please provide a copy of the list of priority diseases, syndromes, and conditions
- 2. Are standard case definitions available for all priority diseases, syndromes, and conditions under national surveillance at all levels?** Yes/No/Don't know
If yes, obtain a copy of these definitions.
- 3. Are predefined action thresholds (e.g., alert and epidemic) established for these priority diseases, syndromes, and conditions under current national surveillance?**
Yes, for all diseases/Yes, for some diseases/No/Don't know
 - a. If yes, are they analyzed on a regular basis (e.g., quarterly, biannually) to triage out deviations from true outbreaks/events?
- 4. Is Annex II of the International Health Regulations (IHR) referenced to determine if an event is notifiable to the World Health Organization (WHO)?**
Yes/No/Don't know

5. Are there any surveillance systems in place that are used in country for early warning surveillance?

Surveillance system type	Name of system(s)	Mechanism for data collection (e.g., paper, electronic [DHIS-2, etc.])	Sectors/ partners involved (e.g., Ministry of Agriculture (MoAg), Non-governmental organizations (NGOs))	Level(s) implemented (e.g., facility, community, etc.)	Any referenced guidance
Event-based Surveillance ¹					
Indicator base surveillance²					
Notifiable disease surveillance (e.g., IDSR, IDS)					
Collaborative surveillance (e.g., IDS)					
Disease specific (e.g., HIV, TB, malaria)					

¹ Event-based surveillance (EBS) is defined by Africa CDC as the organized collection, monitoring, assessment, and interpretation of primarily unstructured ad hoc information regarding health events or risks, which may represent an acute risk to human, animal, plant, or environment health. Also, when referring to EBS in general this can include all modalities (i.e., media scanning, hotline, community, and facility-based).

² Indicator-based surveillance (IBS): The systematic collection, monitoring, analysis, and interpretation of structured health-related data (indicators), produced by health facilities or other defined sources. Reporting is based on standardized case definitions of selected priority diseases or conditions.(ref: <https://www.who.int/publications/i/item/9789240063587>)

Community-based surveillance ³					
Laboratory-based surveillance					
Partner/ NGO run systems					
Other IBS: Syndromic/Sentinel Surveillance					
ILI/SARI					
Acute Febrile Illness (AFI)					
Other syndromic/ sentinel surveillance					
Other:					

³ Community-based surveillance (CBS): The systematic detection and reporting of events of public health significance or of case cases of specific diseases, within a community, by community members. (ref: <https://www.who.int/publications/i/item/9789240063587>)

EBS Implementation

- 6.** Do you have a National EBS technical working group (TWG)? Yes/No
 - a. If yes, who is involved?
- 7.** Has the country/National EBS TWG prepared and agreed upon a list of priority events (signal definitions) for EBS? Yes/In Progress/No
 - a. If yes and/or in progress, please share
- 8.** Have the potential sources of EBS data been mapped out/identified? Yes/In Progress/No
 - a. If yes and/or in progress, please share what these are
- 9.** Is risk assessment part of the routine EBS workflow? Yes/No
 - a. If yes, please share when this is performed and what tool is being used to do this?
 - b. If yes, do you use the risk assessment outcomes to inform alert and response related activities? Yes/No
 - i. If yes, please share an outline of the actions taken for events by risk level.
- 10.** Are there EBS focal points/persons identified at the various levels of EBS implementation? Yes/No
 - a. If yes, how many focal points/persons, at which administrative level (see Annex 2 visual), and what is the local name for this level?
 - i. National
 - ii. Intermediate
 - iii. Facility
 - iv. Local
 - v. Community
 - vi. Other:
 - b. If yes, are there established roles and responsibilities for EBS staff?
- 11.** Do you maintain a list of national or regional subject matter experts (SMEs)/networks that specialize in particular disease areas who might be called to assist with the signal verification and risk assessment? Yes, for all/Yes, for some/No

12. Are routine EBS reports disseminated from the national level? Yes/No
 - a. If yes, how frequently are these produces (daily, weekly, monthly)
 - b. If yes, please provide details as to who is receiving these reports
 - c. If yes, please provide details as to the frequency of EBS reporting
 - d. If yes, please describe how EBS reports are disseminated

13. Are other sectors beyond human health-related such as animal, agriculture, environment, customs, immigration, etc. actively and routinely working with ministry partners in the EBS implementation? Yes/No
 - a. If yes, who and what partners?
 - b. What is the frequency of interactions in working with these partners?

14. Specifically, is there a **community event-based surveillance (CEBS) system** in place to report unusual, unexpected, or new events as they present in the community (e.g., high absenteeism at school, mass poultry die off, etc.)? Yes/No/Don't know
 - a. If No, are you interested in building capacity for this? Yes/No
 - b. If yes,
 - i. Which partners are engaged for community work (e.g., CHW, CAHW, community organizations, IFRC, traditional healers, volunteers, etc.)?
 - ii. Have you developed a community-level signal list? Yes/Yes, but needs refinement/No
 1. If yes, have the signals been translated into the local language where implementation will take place? Yes/No
 2. If yes, were they field tested and CEBS staff trained on these prior to the implementation of EBS? Yes/No
 3. If yes, are there regular team meetings with CEBS staff to discuss new signals detected and ensure appropriate follow-up? Yes/No
 - iii. What are the sources that you use to inform this system (e.g., detect signals, etc.):
 1. Community health workers (CHWs)
 2. Community animal health workers (CAHW)
 3. Other, specify:
 - iv. Do you have specific forms/tools (e.g., signal log) to capture and report CEBS data? Yes/No
 1. If yes, check the type of forms/tools (check all that apply)
 - a. Paper-based
 - b. Electronic based/App

- i. If yes, name of the app_____
 - c. SMS/WhatsApp
 - d. Phone call
 - 2. Do all sub-national levels (e.g., intermediate, local) use the same forms/tools? Yes/No/Don't know
- v. If a signal log is available, specify which type of information is registered:
Check all that apply:
 - ☐ Date/time of signal started
 - ☐ Date/time signal detected
 - ☐ Description of the signal including any symptoms or outcomes and number of people/animals affected
 - ☐ Location of report (specify granularity of admin level)
 - ☐ Source of report including contact information
 - ☐ Others? List here:
- vi. Are EBS communication materials (e.g., posters, infographics) available and distributed at the community level? Yes/No
 - 1. If yes, please upload a picture of materials
- vii. Is a multisectoral, One Health approach used for event detection and reporting? Yes/No/Don't know
- viii. Is CEBS conducted at the borders in collaboration with adjacent countries? Yes/No/Don't know
 - 1. If yes, with who and/or what borders?

15. Specifically, is there a **facility event-based surveillance (FEBS) system (e.g., healthcare facility, animal health care centers, wastewater facilities, etc.) in place to report clusters of unknown illnesses in patients or healthcare workers? Yes/No/Don't know**

- a. If No, are you interested in building capacity for this? And for which facilities?
- b. If Yes,
 - i. What are all the sources of information that are used to inform this system: Check all that apply:
 - ☐ Public hospitals/clinics
 - ☐ Major/minor health centers
 - ☐ Local/village health services
 - ☐ Private sector
 - ☐ Private clinics
 - ☐ Public animal health facilities
 - ☐ Private animal health facilities

- ☐ Laboratories
 - ☐ Wastewater facilities
 - ☐ Other, specify:
- ii. Have you developed a facility-level signal list for each facility type?
Yes/Yes, partial OR needs refinement/No
 1. If yes, were they field tested and FEBS staff trained prior to the implementation of EBS? Yes/No
 2. If yes, are there regular team meetings with FEBS staff to discuss new signals detected and ensure appropriate follow-up? Yes/No
- iii. Do you have specific forms/tools to capture and report FEBS data?
Yes/No
 1. If yes, check the type of forms/tools (check all that apply)
 - a. Paper-based
 - b. Electronic based/App
 - i. If yes, name of the app_____
 - c. SMS/WhatsApp
 - d. Phone call
- iv. Are FEBS communication materials (e.g., posters, “bench” buddies, etc.) available and distributed throughout the involved facilities? Yes/No
 1. If yes, please upload a picture of materials
- v. Is a multisectoral, One Health approach used for event detection and reporting? Yes/No/Don’t know

16. Specifically, is **media scanning** currently being conducted to detect unusual, unexpected, or new health events from media sources? Yes/No/Don’t know

- a. If No, are you interested in building capacity for this?
- b. If Yes:
 - i. Have you developed a signal list? Yes/Yes, but needs refinement/No
 1. If yes, were they field tested, and staff trained prior to the implementation of EBS? Yes/No
 2. If yes, are there regular team meetings with Media Scanning staff to discuss new signals detected and ensure appropriate follow-up? Yes/No
 - ii. What are all the sources that you use to inform this system: Check all that apply:
 - ☐ Printed media reports
 - ☐ Electronic (e.g., internet-based) media reports

- ☐ TV/Radio broadcasts
 - ☐ Social media platforms
 - ☐ A hotline or call center
 - ☐ WhatsApp messaging
 - ☐ SMS messaging
 - ☐ Other, specify:
- iii. If electronic (e.g., internet-based) media reports are being used as a source, are media aggregator platforms being used to support this? If so which ones:
1. Epidemic Intelligence from Open Sources (EIOS)
 2. GPHIN
 3. Pro-Med
 4. Health Map
 5. Other:
- iv. If media aggregator platforms (e.g., EIOS) are not being used, what obstacles are preventing the adoption of these for media scanning?
- v. Is there a checklist of sources to guide media scanning? Yes/No
- vi. At what level(s) is the system implemented?
- ☐ National
 - ☐ Intermediate
 - ☐ Facility
 - ☐ Local
 - ☐ Community
 - ☐ Other, specify:
- vii. Do you have specific forms/tools to capture and report media scanning data? Yes/No
1. If yes, type of forms/tools
 - a. Paper-based
 - b. Electronic based
- viii. Is a multisectoral, One Health approach used for event detection and reporting? Yes/No

- 17.** Specifically, is there a **hotline system or call center** or other mechanism in place for rapid reporting of unusual, unexpected, or new health events from the community? Yes/No
- a. If no, are you interested in building capacity for this?
 - b. If yes,

- i. How does the public report information into the system (e.g., what is the number associated with the hotline/call center, etc.)?
- ii. Have you developed a signal list? Yes/Yes, but needs refinement/No
 1. If yes, were they field tested, and staff trained prior to the implementation of EBS?
 2. If yes, are there regular team meetings with hotline/call center staff to discuss new signals detected and ensure appropriate follow-up? Yes/No
- iii. What are the sources that you use to inform this system:
 - ☐ A hotline or call center
 - ☐ WhatsApp messaging
 - ☐ SMS messaging
 - ☐ Social media platforms
 - ☐ Other, specify:
- iv. At what level(s) is the system implemented?
 - ☐ National
 - ☐ Intermediate
 - ☐ Facility
 - ☐ Local
 - ☐ Community
 - ☐ Other, specify:
- v. Do you have specific forms/tools to capture, manage, and report hotline/call center data? Yes/No
 1. If yes, type of forms/tools (check all that apply)
 - a. Paper-based
 - b. Electronic based
 - i. If electronic, name of the system_____
 - c. Interactive voice response (IVR)⁴
- vi. If IVR is used, what system is used?
- vii. Is a multisectoral, One Health approach used for event detection and reporting? Yes/No

⁴ Interactive voice response, or IVR, is an automated telephone system that combines pre-recorded messages or text-to-speech technology with a dual-tone multi-frequency (DTMF) and speech recognition interface to engage callers, allowing them to provide and access information without a live agent. This kind of function can support triaging of hotline calls not relevant to EBS.

Information systems

18. Does the country have a centralized electronic event management system (EMS)?

Yes/No

a. If Yes,

- i. What platform/software is the system built? _____
- ii. Is the ministry receiving technical support externally in implementing this system? Yes/No/Don't know
 1. If yes, specify the technical partner ____
- iii. At what level(s) is the EMS implemented?
 1. National
 2. Intermediate
 3. Facility
 4. Local
 5. Community
 6. Other, specify:
- iv. Is it interoperable with the national IBS systems (e.g., DHIS-2, SORMAS, other)? Yes/No
 1. If yes, describe which systems and how the systems share data:
- v. Is it interoperable with other country systems (e.g., emergency response system, LIMS, point of entry, etc.)? Yes/No/Don't know
 1. If yes, describe which systems and how the systems share data:
- vi. Is it interoperable with other sector systems (e.g., One Health, animal health, environment, etc.)? Yes/No
 1. If yes, describe which systems and how the systems share data:
- vii. How do the sub-national jurisdictions report to you regularly? If more than one method is used, specify the level and EBS type for each method used.
 1. Enter direct into an EMS
 2. Enter data into an electronic tool, specify tool:
 3. Send a soft copy of an aggregated report
 4. Send a hard copy of an aggregated report
 5. Send a soft copy of the collection form/log
 6. Send a hard copy of the collection form/log
 7. Phone call
 8. SMS/WhatsApp
 9. Other, specify:

- viii. Does the EMS include an aspect of routine monitoring and/or collect indicator data that can help monitor and evaluate surveillance performance and timeliness? Yes/No
 - 1. If yes, please share what indicators are being monitored
- ix. What workstreams does the EMS support? (check all that apply):
 - ☐ Data entry
 - ☐ Signal verification
 - ☐ Risk assessment
 - ☐ Data analysis
 - ☐ Data visualization
 - ☐ Report generation
 - ☐ Alert notification
 - ☐ Response monitoring (e.g., staff deployment, response actions)
 - ☐ Others? List here:
- b. If there is no EMS and/or the EMS is not implemented sub-nationally, how are data currently managed, stored, and shared/reported in a systematic way at all levels of EBS by type of EBS implemented:
 - i. National
 - ii. Intermediate
 - iii. Facility
 - iv. Local
 - v. Community
 - vi. Other, specify:

19. What type(s) of software tools are utilized for analyzing and/or interpreting surveillance data (e.g., Epi Info, Excel, SAS, STATA, R, Tableau, Microsoft BI)?

20. What equipment do you have to assist with data collection and storage at all levels of EBS?

Equipment	National	Intermediate	Facility	Local	Community
Computer	Available? Yes/No	Available? Yes/No	Available? Yes/No	Available? Yes/No	Available? Yes/No
	Est. Number ____	Est. Number ____	Est. Number ____	Est. Number ____	Est. Number ____
	Is this adequate? Yes/No	Is this adequate? Yes/No	Is this adequate? Yes/No	Is this adequate? Yes/No	Is this adequate? Yes/No
Tablets	Available? Yes/No	Available? Yes/No	Available? Yes/No	Available? Yes/No	Available? Yes/No
	Est. Number ____	Est. Number ____	Est. Number ____	Est. Number ____	Est. Number ____
	Is this adequate? Yes/No	Is this adequate? Yes/No	Is this adequate? Yes/No	Is this adequate? Yes/No	Is this adequate? Yes/No
	Is connectivity adequate for all/some sources? Yes/No	Is connectivity adequate for all/some sources? Yes/No	Is connectivity adequate for all/some sources? Yes/No	Is connectivity adequate for all/some sources? Yes/No	Is connectivity adequate for all/some sources? Yes/No
Cell phones	Available? Yes/No	Available? Yes/No	Available? Yes/No	Available? Yes/No	Available? Yes/No
	Est. Number ____	Est. Number ____	Est. Number ____	Est. Number ____	Est. Number ____
	Is this adequate? Yes/No	Is this adequate? Yes/No	Is this adequate? Yes/No	Is this adequate? Yes/No	Is this adequate? Yes/No
	Is connectivity adequate for all/some sources? Yes/No	Is connectivity adequate for all/some sources? Yes/No	Is connectivity adequate for all/some sources? Yes/No	Is connectivity adequate for all/some sources? Yes/No	Is connectivity adequate for all/some sources? Yes/No
Other relevant equipment Specify	Equipment: ____	Equipment: ____	Equipment: ____	Equipment: ____	Equipment: ____
	Available? Yes/No	Available? Yes/No	Available? Yes/No	Available? Yes/No	Available? Yes/No
	Est. Number ____	Est. Number ____	Est. Number ____	Est. Number ____	Est. Number ____
	Is this adequate? Yes/No	Is this adequate? Yes/No	Is this adequate? Yes/No	Is this adequate? Yes/No	Is this adequate? Yes/No

Preparedness and Response

21. When an alert is made for an event, who is notified and how is this performed?

- a. within the MoH/NPHI?
- b. within other relevant sectors:
- c. for multisectoral events: (also include any relevant sectors [e.g., MoAg], partners, etc.)

22. Describe who is responsible for managing and responding to events?

- a. within the MoH/NPHI (e.g., National Emergency Operations Center (EOC)):
- b. within the other sectors (e.g., EOC, Epidemic Intelligence hub):
- c. for multi-sectoral events (e.g., EOC, Multisectoral Coordination Mechanism):

23. How are those responsible for managing and responding to events alerted of events detected through EBS?

- a. within the MoH/NPHI:
- b. within the other sectors involved in EBS:
- c. for multi-sectoral events:

Laboratory

24. Is there national laboratory capacity in place for:

- a. Diagnosis and confirmation of priority pathogens in humans: Yes/No
- b. Diagnosis and confirmation of priority pathogens in animals: Yes/No
- c. Identification of antimicrobial resistance: Yes/No
- d. Identification of environmental contamination or toxins: Yes/No
- e. Whole genome sequencing: Yes/No

25. Is the EBS work informed by laboratory results? Yes/No

- a. If yes, how?
 - i. To help identify/confirm cause of unknown events
 - ii. As part of our facility based EBS implementation (see 6.b.ix)
 - iii. Other, please explain:

Plans, Assessments, and Research

26. Does the country have a strategic plan in place for surveillance that is inclusive of

EBS? Yes/No/Don't know

- a. If yes, can you share and/or upload this plan? Yes/No
- b. If not, is this needed?

27. Are there SOPs or guidelines in place for all EBS types/modalities established?

Yes, all/Yes, partial/No

- a. If partial, which ones?
- b. If yes
 - i. Do these include the flow for EBS information, reporting and feedback? Yes/No
 - ii. Do these cover multisectoral, One Health implementation (e.g., animal, agriculture, environment, customs, immigration etc.)? Yes/No
 - iii. Do these cover cross-border implementation? Yes/No

28. Has the government developed an implementation plan for EBS in the country?

Yes/No

- a. If yes, can you share and/or upload this plan? Yes/No
- b. If not, is this needed?

29. Does the country have a monitoring and evaluation plan in place for EBS? Yes/No

- a. If no, is this needed? If not, is support needed to develop?
- b. If yes,
 - i. Please share a copy of this, including the list of indicators
 - ii. Are 7-1-7 indicators included in the list of M&E indicators? Yes/No/Don't know
 - 1. If no, is this needed?
- c. What is the timeframe (in days or hours) used within which signals should be verified after detection?
- d. What is the timeframe (in days or hours) used within which events should undergo a risk assessment after being verified?
- e. What is the timeframe (in days or hours) used within which events should be reported and responded to after detection?

30. Have you performed any evaluations of the EBS system? Yes/No/Don't know

- a. If yes,
 - i. How frequently do you evaluate the system?
 - ii. When was the last evaluation conducted?
 - iii. What were some of the primary outcomes?
 - iv. Do you have an evaluation report or other publication on the outcomes?
Yes/No
 - 1. If yes, - upload
- b. If no, is an evaluation needed?

31. Have you identified any gaps in effectively implementing EBS at the country level? Yes/No

- a. If yes - in what areas? (Check all that apply)
 - i. Development of SOPs and implementation plans
 - ii. Training of TOTs for EBS staff, including refresher trainings
 - iii. Cascaded training, including refresher trainings
 - iv. Development of signals and priority events
 - v. Understanding sensitivity of existing signals
 - vi. Guidance on risk assessment
 - vii. Mentorship or supportive supervision
 - viii. Inability to determine impact
 - ix. Others, specify:

32. Does the NPHI/MoH have a public health research agenda inclusive of EBS?

Yes/No

33. Does the country perform any in-depth analysis on the data collected from the EBS? Yes/No

- a. If yes, please upload the most recent EBS report
- b. If yes, please upload any publications
- c. Other, specify:

Legislation

34. Is there a legal or policy framework for establishing the EBS system in the country? Yes/No

- a. If yes, please upload
- b. If no, is this needed?

35. Have coordination mechanisms between various levels of government been put in place for integration, sharing and use of EBS data at all levels? Yes/No

- a. If yes, please upload
- b. If no, is this needed?

36. Have multisectoral coordination mechanisms across relevant collaborating sectors been established to including integration, sharing and use of EBS data?

Yes/No

- a. If yes, please upload
- b. If no, is this needed?

37. Are there regional or bilateral arrangements with neighboring countries to share/report EBS data? Yes/No

- a. If yes, what countries?
- b. If yes, please upload current agreements
- c. If no, are these needed?

Workforce

38. How many staff are currently trained to implement EBS in the country by level and sector:

- a. National
- b. Intermediate
- c. Facility
- d. Local
- e. Community
- f. Other, specify:

39. Does the NPHI/MoH have a workforce strategy in place that is inclusive of EBS?

Yes/No

40. Is there a training curriculum and training plan in place for EBS? Yes/No

- a. If yes,
 - i. Which modalities does it cover?
 - ☐ CEBS
 - ☐ FEBS
 - ☐ Hotline
 - ☐ Media scanning
 - ii. How often does training occur (by administrative level and EBS type)?
 - iii. When were these materials last reviewed and updated?
 - iv. Is there an element of TOT
- b. If yes, please upload
- c. If no, is this needed?

41. Are there regular EBS monitoring and supportive supervision visits? Yes/No

- a. If yes, how often are supportive supervision visits performed?
- b. Do supportive supervision visits differ by level within the government? Yes/No
- c. If yes, please describe how these visits differ?

Structure/Infrastructure

42. Is there a National Public Health Institute (NPHI) or a unit/department serving the functions of a NPHI within the MoH? Yes/No

If yes,

- a. How are roles and responsibilities for EWAR implementation allocated across the NPHI and MoH?
- b. Are data readily and effectively shared between the NPHI and the MoH? Yes/No

43. Is there an EOC or EI hub and where are these administratively situated (e.g., within the MoH, NPHI, other)?

44. If EBS implementation is implemented across two different organizational groups within an MoH or NPHI:

- a. Who (e.g., department) is responsible for implementing each of the following:

- i. IBS
 - ii. Surveillance aspects of EBS
 - iii. Response aspects of IBS and EBS
- b. How are data shared between the groups and is this effective?
- c. Is the same EMS or data management system used? Why/why not?

45. Is the EBS reporting and feedback structure (e.g., flow of information) aligned to the national IBS reporting structures? Yes/No

- a. If not, is this needed?

46. Does the group coordinating EBS have trouble maintaining reliable internet access? Yes/No

- a. If yes, how often is internet access disrupted?
 - i. Hourly
 - ii. Daily
 - iii. Weekly
 - iv. Monthly
 - v. Other, specify:

47. Is access to a phone network available 24/7? Yes/No

- a. If yes,
 - i. What is the mobile phone network coverage in the country?
 - ii. What is the level of mobile access and ownership by community members?
 - iii. Is there a map of available phone network coverage for the country?
Yes/No
 - 1. If yes, please upload
- b. If no, describe the alternative means of communication/access.

48. Is electricity or power shortage a common challenge? Yes/No

- a. If yes, how often is power disrupted?
 - i. Hourly
 - ii. Daily
 - iii. Weekly
 - iv. Monthly
- b. If yes, are there any available back-up generators in use? Yes/No
 - i. If yes, where?

Finance

49. Is there an annual budget for the NPHI/MoH/Country to discharge its functions?

Yes/No

50. Is there a dedicated line item to implement EBS in the country budget? Yes/No

a. If yes, what components of EBS are currently funded under this national budget? Check all that apply.

- ☐ Staffing
- ☐ Capacity building (e.g., training)
- ☐ Vehicles running to support verification and/or investigation
- ☐ Information and Communication Technology (ICT) (e.g., computers, printers, cell phones/other communication devices, etc.)
- ☐ Other, specify:

b. What are the areas or components of EBS that are left uncovered or underfunded?

51. Who are the other financiers of the EBS system? Check all that apply.

- ☐ Regional organizations
- ☐ Development Partners
- ☐ Private sector
- ☐ Other sources (specify):

52. Has there been support obtained from the US CDC (or another group) to facilitate EBS implementation in the country for the following? Check all that apply

- a. Field teams (travel costs)
- b. Materials, supplies and reagents (incl. PPE etc.)
- c. Communication and logistics (printing, transport etc.)
- d. Other known financial support (e.g., cooperative agreements, in-kind donations)
- e. In addition to the US CDC, what other organizations have financially supported EBS implementation? (specify)

Conclusion

53. Are there any areas where you identify additional support from US CDC? (Check all that apply)

- i. Development of SOPS and implementation plans
- ii. Training materials
- iii. Development of signals and priority events
- iv. Signal sensitivity analysis
- v. System evaluation
- vi. Guidance on risk assessment
- vii. Mentorship or supportive supervision
- viii. Inability to determine impact
- ix. Others, specify:

54. Any other comments you would like to share?

----End of assessment----

Annex 1: Glossary & Definitions

7-1-7: a metric, proposed by Resolve to Save Lives that suggests there be 7 days to detect a suspected infectious disease outbreak, 1 day to notify public health authorities to start an investigation and 7 days to complete an initial response.

Alert: A public health signal that has been i) verified to be an event; ii) risk assessed; and iii) requires an intervention (an investigation, a response or a communication with partners or the public). (See also signal and event).

Alert management: The systematic process of managing all incoming information from signal verification to risk assessment and characterization to decide if a response is required to mitigate the public health risk. (See also verification and risk assessment).

Alert threshold: A predefined number of cases (or, e.g., proportion, rate, trend). Once an alert threshold is reached or surpassed, a signal is triggered, and the process of verification and risk assessment and risk characterization is started.

All-hazard: The all-hazard approach acknowledges that while hazards vary in source (natural, technological, societal), they often challenge health systems in similar ways and demand a multisectoral response. Thus, risk reduction, emergency preparedness, response actions and community recovery activities are usually implemented using the same model, regardless of cause.

Community event-based surveillance (CEBS): Africa CDC defines community event-based surveillance as the detection and reporting of unusual health events or health risks occurring within a community, by community members including community volunteers, community health or animal health workers, the public, religious leaders, civil society members, teachers, and other similar groups.

Community animal health worker (CAHW): Defined by WOAHA as a person selected by their own community and provided with short, initial, or recurring vocational training to perform basic animal health and animal husbandry-related tasks, who is accountable to a veterinary para-professional and/or veterinarian, and who is currently active in their

community. The CAHW can also play an important role in a range of sanitary tasks such as disease reporting.⁵

Community health worker (CHW): CHWs provide health education and referrals for a wide range of services, and provide support and assistance to communities, families and individuals with preventive health measures and gaining access to appropriate curative health and social services. They create a bridge between providers of health, social and community services and communities that may have difficulty in accessing these services. CHWs may also be known as community health volunteers, among other names. According to a WHO Study Group, community health workers may be members of the communities where they work, should be selected by the communities, are answerable to the communities for their activities, and should be supported by the surveillance and/or health system but not necessarily a part of its organization.

Early Warning and Response (EWAR): Early Warning Alert and Response (EWAR) is defined as the organized mechanism to rapidly detect and respond to signals that might indicate potential acute public health events.

Emergency Operations Center (EOC): WHO defines an EOC as a physical location for the coordination of information and resources to support incident management activities. Such a center may be a temporary facility, or may be established in a permanent location.

Epidemic: see Outbreak

Epidemic Intelligence: The systematic collection, analysis, and communication of information from different sources to detect, verify, assess, and investigate events and infectious disease risks with the objective of providing early warning.

Epidemic Intelligence from Open Sources (EIOS): Web-based, media aggregating platform that allows users to rapidly sift through huge volumes of open-source information from a wide variety of sources in order to help identify public health events. EIOS is a fit-for-purpose but constantly evolving web-based system designed to augment and accelerate global public health intelligence (PHI) activities, built on a long-standing

⁵ The World Organisation for Animal Health (WOAH) does not have an official definition for CAHW. However, in the context of activities of the Capacity Building Department at WOAH regarding veterinary workforce development, this informal, unofficial definition is being used to communicate the meaning of CAHW to distinguish CAHWs from veterinary paraprofessionals.

collaboration between WHO and the Joint Research Centre (JRC) of the European Commission.

Evaluation: The periodic assessment of the relevance, effectiveness, and impact of activities in the light of the objectives of the surveillance and response systems.

Event: The International Health Regulations (IHR) define an event as “[...] *a manifestation of disease or an occurrence that creates a potential for disease; [...]*” (which can include events that are infectious, zoonotic, food safety, chemical, radiological, or nuclear in origin and whether transmitted by persons, vectors, animals, goods/food, or through the environment.).

Event-based surveillance (EBS): Defined by Africa CDC as the organized collection, monitoring, assessment, and interpretation of primarily unstructured ad hoc information regarding health events or risks, which may represent an acute risk to human, animal, plant, or environment health.

Facility event-based surveillance (FEBS): Event-based surveillance that is conducted in a facility (e.g., hospitals, clinics, animal health clinics, laboratories, water treatment facilities, educational facilities, etc.). EBS focal points (FP) identified at these facilities support the detection and reporting of signals or events happening at these facilities that are not covered through routine indicator-based surveillance.

Healthcare facility event-based surveillance (HEBS): Event-based surveillance that is conducted in hospitals and healthcare facilities. Healthcare workers are involved as primary reporting sources, such as during patient consultations, or as secondary sources reporting unusual health events or health risks picked up through patient consultations.

Hotline: A hotline (toll-free) is a phone line that the general public can use to contact an institution/organization about a particular health concern.

Indicator-based surveillance (IBS): The systematic collection, monitoring, analysis, and interpretation of structured health-related data (indicators), produced by health facilities or other defined sources. Reporting is based on standardized case definitions of selected priority diseases or conditions.

Interactive voice response (IVR): is an automated telephone system that combines pre-recorded messages or text-to-speech technology with a dual-tone multi-frequency (DTMF) and speech recognition interface to engage callers, allowing them to provide and access information without a live agent. This kind of function can support triaging of hotline calls not relevant to EBS.

Intermediate administrative level: Intermediate administrative levels may be defined differently in different countries. For this document, the intermediate level is the health administrative level(s) below the national-level that is responsible for conducting preliminary investigations and implementing responses to reported health-related events or suspected outbreaks in a given jurisdiction. The intermediate level may otherwise be referred to as districts or counties, among others. Some countries have two administrative layers (e.g., provincial and district) that make up their intermediate level.

Local administrative level: Local administrative levels may be defined differently in different countries. For this document, a local administrative level is the lowest administrative division within a country, directly above the community-level.

Media scanning (also known as “media monitoring”): The active monitoring of the content of media sources on a continuing basis to get information about specific topics. In the context of EBS this involves the systematic monitoring of information available in local, national, and international media sources including the internet and social media networks to detect unusual health events.

Monitoring: Monitoring in the context of surveillance and response systems refers to the routine and continuous tracking of the implementation of planned surveillance activities, and of the overall performance of these systems.

Multisectoral: Participation of more than one sector working together with a common vision and perspective on a joint program or response to an event (e.g., a joint investigation by public health, animal health, education, and law enforcement).

Multisectoral, One Health coordination mechanism (MCM): A multisectoral, One Health coordination mechanism (MCM) refers to any formalized, standing, group that acts to strengthen or develop collaboration, communication, and coordination across the sectors responsible for addressing health concerns at the human-animal-environment interface. An MCM has routine, ongoing functions and is responsible for coordination,

leadership, and governance of efforts among the relevant sectors to achieve jointly determined and agreed common goals.

One Health: One Health is an integrated, unifying approach that aims to sustainably balance and optimize the health of people, animals, and ecosystems. It recognizes the health of humans, domestic and wild animals, plants, and the wider environment (including ecosystems) are closely linked and interdependent. The approach mobilizes multiple sectors, disciplines, and communities at varying levels of society to work together to foster well-being and tackle threats to health and ecosystems, while addressing the collective need for clean water, energy and air, safe and nutritious food, taking action on climate change, and contributing to sustainable development.⁶

National Public Health Institute (NPHI): defined by IANPHI (the International Association of Public Health Institutes) as a government agency, or closely networked group of agencies, that provides science-based leadership, expertise, and coordination for a country's public health activities.

Outbreak (synonym Epidemic): A disease outbreak is the occurrence of cases of disease more than what would normally be expected in a defined community, geographical area, or season. An outbreak may occur in a restricted geographical area or may extend over several countries. It may last for a few days or weeks, or for several years. A single case of a communicable disease long absent from a population, or caused by an agent (e.g., bacterium or virus) not previously recognized in that community or area, or the emergence of a previously unknown disease, may also constitute an outbreak and should be reported and investigated.

Public Health Intelligence: WHO defines this as the process of monitoring global health threats by gathering and monitoring information on public health events via open source, governmental, and other domains of intelligence. (See also epidemic intelligence)

Public Health Risk: The likelihood of an event that may adversely affect the health and livelihoods of human populations. (See also risk)

Reporting: The process by which signals, or events are brought to the knowledge of the health authorities.

⁶ Quadripartite, One Health High Level Expert Panel's definition of "One Health".

Response: Any public health action triggered by the detection of a public health risk (e.g., monitoring of the *event*, information of the public, triggering field investigation and/or implementation of any control or mitigation measures). The nature of the response will have to be adapted according to the nature of the public health risk.

Risk: Africa CDC defines this as the likelihood of an event resulting in negative consequences for health (e.g., animal health, public health, etc.). (See also public health risk)

Risk assessment: A systematic process for gathering, assessing, and documenting information to assign a level of risk to an event. Data collected through the risk assessment process are used to inform risk characterization and any immediate actions to be taken in response.

Risk characterization: According to WHO, once a risk assessment team has carried out hazard, exposure, and context assessments of an event, a level of risk should be assigned. This process is called risk characterization.

Rumors: Unverified information regarding disease occurrence received from informal sources.

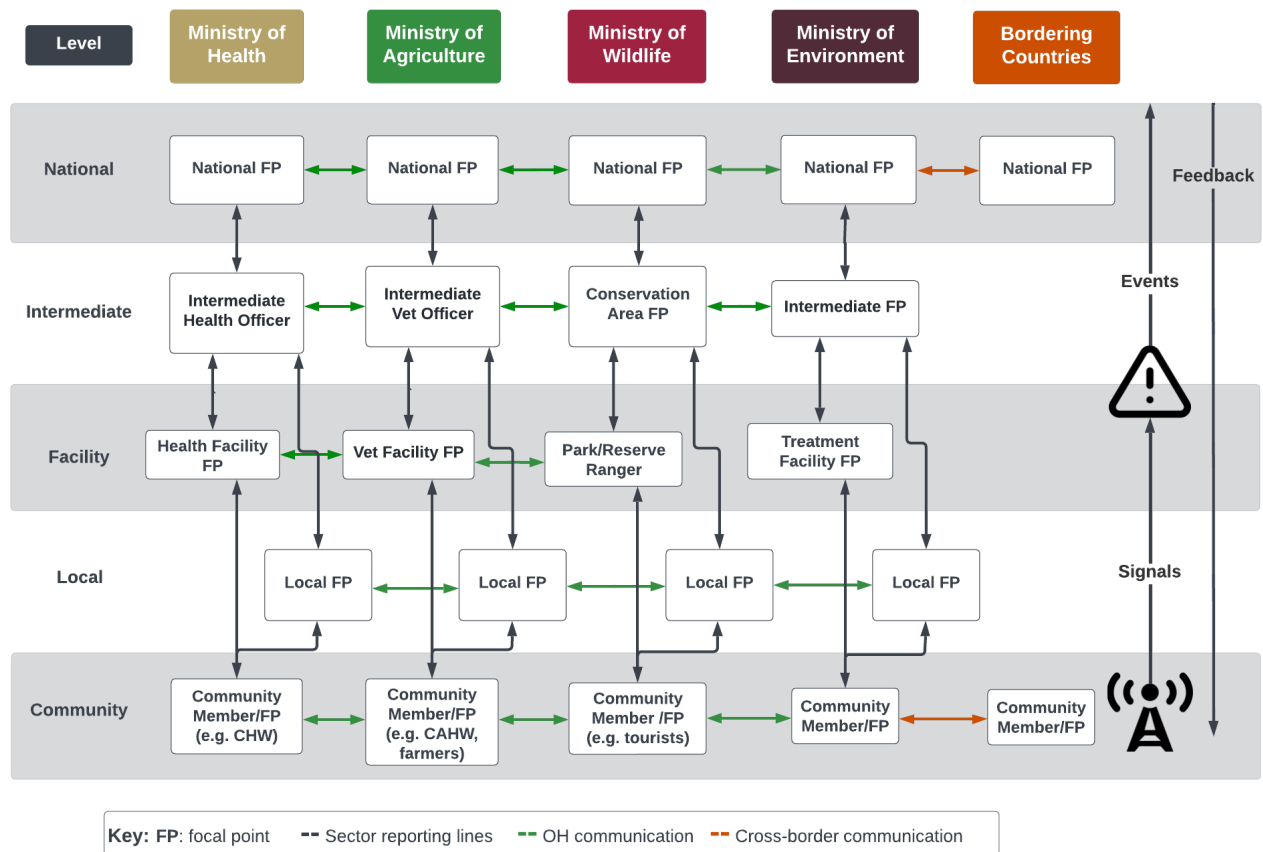
Signals: The initial detection (by IBS or EBS) of a potential public health event, prior to verification. Signals may consist of information/reports of cases or deaths (individual or aggregated), potential exposure of human beings to biological, chemical, or radionuclear agents, or of the occurrence of natural or manmade disasters. Once identified, signals must be verified. When it has been verified, a signal becomes an “event”.

Triage: The process of screening and selecting information that is relevant for early detection purposes. The process of triaging involves two steps: 1) filtering, or screening out, irrelevant or duplicative information; and 2) selection, which is a human analyst driven process that includes selecting relevant reports based on the focus of a particular EBS unit and/or country priority. Once information is triaged, it becomes a “signal”.

Verification: The proactive assessment by EWAR of the validity of the signals collected – eliminating hoaxes, false rumors, and artefacts. This may involve contacting the primary source or additional sources or performing field investigations.

Zoonotic disease or zoonoses: an infectious disease that can be shared between animals and humans; can be spread by food, water, fomites, or vectors.

Annex 2: Example multisectoral reporting structure



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